

Notified Body
TÜV Rheinland
LGA Products GmbH

Tillystraße 2
90431 Nürnberg

notified by the
Bundesnetzagentur für Elektrizität, Gas,
Telekommunikation, Post und Eisenbahnen

under No. 0197

herewith issues an

EU-Type Examination Certificate

within the meaning of Annex III Module B of the 2014/53/EU Radio Equipment Directive (RED)
for compliance with the essential requirements of this directive

Registration Number: RT 60175036 0001

Evaluation Report Nr.: CN247K7S 001

Manufacturer:

Beijing InHand Networks Technology
Co., Ltd.
Room 501,
floor 5, building 3, yard 18, ziyue road, chaoyang district,
100020 Beijing
P.R. China

Product:

Radio Equipment
(Industrial Cellular Router)

Type
Identification:

IR315 IR3X5 (X=0-9)
(InHand)

Essential
requirements:

2014/53/EU (RED)
Article 3.1a Health
Article 3.1a Electrical Safety
Article 3.1b EMC
Article 3.2 Radio spectrum

The technical design of the assessed type has been verified based on the technical documentation presented by the manufacturer according to Annex III Module B of the Directive. As far as the essential requirements indicated, the Notified Body of TÜV Rheinland LGA Products GmbH confirms, that the technical design of the apparatus meets the essential requirements of the Directive 2014/53/EU Article 3.

This certificate consists of this page and Annex I.

Validity of the certificate is specified in the Annex I.

Date **08.03.2024**



Notified Body

Sam Lin

Equipment

Product : Industrial Cellular Router
Trademark : 
Identification : IR315, IR3X5(Where X represents the numbers 0 - 9)
Product description : The product is a Industrial Cellular Router with 2.4GHz Wi-Fi, GSM, WCDMA and LTE wireless functions.

System description

Frequency band(s) of operation : E-GSM 900, DCS 1800, WCDMA Band I/VIII
LTE Band 1/3/7/8/20/28/38/40/41, 2400-2483.5MHz
Operating frequency : E-GSM 900: Uplink: 880-915MHz, Downlink: 925-960MHz
DCS 1800: Uplink: 1710-1785MHz, Downlink: 1805-1880MHz
WCDMA Band I: Uplink: 1920-1980MHz, Downlink: 2110-2170MHz
WCDMA Band VIII: Uplink: 880-915MHz, Downlink: 925-960MHz
LTE Band 1: Uplink: 1920-1980MHz, Downlink: 2110-2170MHz
LTE Band 3: Uplink: 1710-1785MHz, Downlink: 1805-1880MHz
LTE Band 7: Uplink: 2500-2570MHz, Downlink: 2620-2690MHz
LTE Band 8: Uplink: 880-915MHz, Downlink: 925-960MHz
LTE Band 20: Uplink: 832-862MHz, Downlink: 791-821MHz
LTE Band 28: Uplink: 703-748MHz, Downlink: 758-803MHz
LTE Band 38: Uplink: 2570-2620MHz, Downlink: 2570-2620MHz
LTE Band 40: Uplink: 2300-2400MHz, Downlink: 2300-2400MHz
LTE Band 41: Uplink: 2535-2675MHz, Downlink: 2535-2675MHz
2.4GHz Wi-Fi: 2412-2472MHz
Channel spacing / bandwidth : 200kHz, 1.4MHz, 3MHz, 5MHz, 10MHz, 15MHz, 20MHz, 40MHz
RF output power : E-GSM 900: 33dBm±2.5dB
DCS 1800: 30dBm±2.5dB
WCDMA: 24dBm+1.7dB/-3.7dB
LTE: 23dBm+2.0dB/-2.0dB
2.4GHz Wi-Fi: 15.93dBm (max. e.i.r.p.)
Type of modulation : GMSK, 8PSK, QPSK, 16QAM, DSSS (DBPSK, DQPSK, CCK),
OFDM (BPSK, QPSK, 16QAM, 64QAM)
Type of antenna : External Antenna
Mode of operation (simplex / duplex) : Duplex
Duty cycle (access protocol, if applicable) : Up to 100%
Hardware version : V1.1
Software version : V1.0

Documentation

User information and installation instructions ☒
Block diagram ☒
Circuit diagram ☒
Part list ☒
PCB layout ☒
Photo documentation ☒
Versions of firmware/software used ☒
Statement of compliance with art. 10.2 it can be operated in at least one Member State without infringing applicable requirements on the use of radio spectrum. ☒
Risk Analysis ☒

Conformity Assessment

Applied harmonised standards (Referred to the publication of harmonised standards in the official Journal of the EU at the time of issuance)			
Article	Standard	Test Report No.	Issued by
3.1a Health			
3.1a Safety			
3.1b EMC	EN 301 489-52 V1.2.1	CN24UU57 003	TÜV Rheinland (Shenzhen) Co., Ltd.
3.2 Radio	EN 300 328 V2.2.2	CN24UU57 001	TÜV Rheinland (Shenzhen) Co., Ltd.
	EN 301 511 V12.5.1	CN24UU57 002	

	EN 301 908-1 V15.1.1 EN 301 908-1 V15.2.1 EN 301 908-2 V13.1.1 EN 301 908-13 V13.1.1	R2110A0929-R1 R2110A0929-R5 R2110A0929-R6	TA Technology (Shanghai) Co., Ltd.
3.3 Others			

Applied non-harmonised standards			
Article	Standard	Test Report No.	Issued by
3.1a Health	EN IEC 62311:2020	CN24UU57 002	TÜV Rheinland (Shenzhen) Co., Ltd.
3.1a Safety	EN 62368-1:2014+A11:2017	CN242X1H 001	TÜV Rheinland (Shenzhen) Co., Ltd.
3.1b EMC	EN 301 489-1 V2.2.3 EN 301 489-17 V3.2.4 EN 55032:2015+A11:2020 EN 55035:2017+A11:2020 EN IEC 61000-3-2:2019+A1:2021 EN 61000-3-3:2013+A2:2021	CN24UU57 003	TÜV Rheinland (Shenzhen) Co., Ltd.
3.2 Radio			
3.3 Others			

Other solutions, adopted to meet the essential requirements			
Article	Standard / Solution	Test Report No.	Issued by
3.1a Health			

Rationale for applied non-harmonised standards or other solutions:

- Standards are either in the EMC/LVD OJ, on CENELEC and ETSI website as current, or valid for RED compliance.

Remarks:

- This Type Examination Certificate does not imply assessment of the production of the product and does not permit the use of a TÜV Rheinland mark of conformity.
- This Type Examination Certificate only relates to the assessment of technical documentation to verify that the technical design of radio equipment meets the essential requirements of the RED 2014/53/EU and will not show compliance with essential requirements of other possible applicable EU Directives.
- The manufacturer has declared in compliance with art. 10.2 that the Radio Equipment can be operated in at least one Member State without infringing applicable requirements on the use of radio spectrum.
- Validity of this Type Examination Certificate is limited to the versions of the applied standard. If versions of standards change or modifications are made to the product, this Certificate will be invalidated.